

Adaptive Switching Strategies for the Trisection-False Position Hybrid

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Recap: Where We Started

THE BASE PAPER

Badr, Almotairi & El Ghamry, Mathematics 9 (2021) 1306

Static trisection + false position (TF). Every step evaluates both trisection points x_1 , x_2 and the false-position point x_F , keeps the smallest $|f(x)|$ and narrows the bracket.

Reported: 98 total iterations vs 121 for the bisection–false position hybrid (BF) on 15 equations.

The gap: iterations were compared, but TF pays up to 3 new function calls per iteration, whatever the function is doing.

WHAT WE MEASURE FOR EVERY METHOD

Iterations

f calls

Time

Success

RESEARCH QUESTION

Does adaptivity actually pay, and which signal is the right one to adapt on?

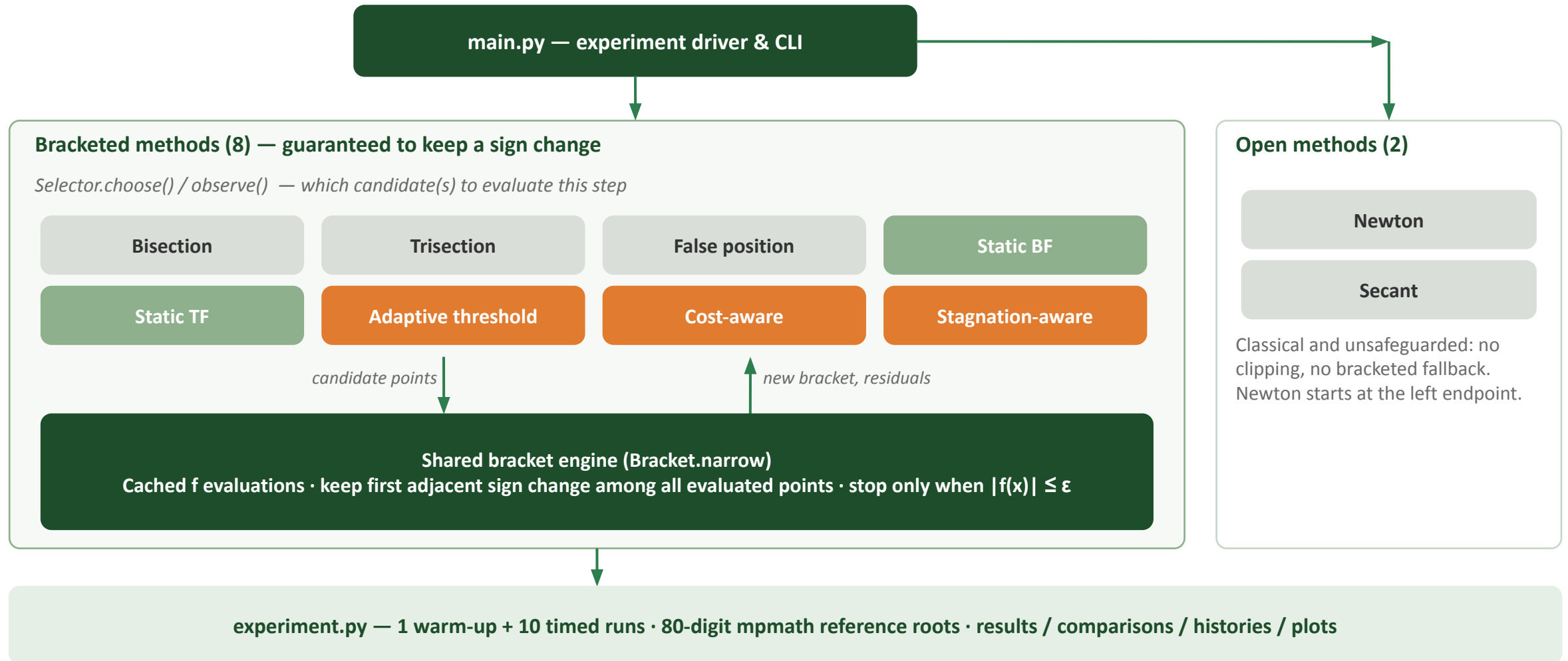
Three adaptive policies replace only the selection rule

Adaptive threshold adapts on residual ratio

Cost-aware adapts on progress per f call

Stagnation-aware adapts on endpoint reuse

Implementation: One Engine, Pluggable Selectors



The Three Policies, As Implemented

1 Adaptive Threshold

SIGNAL

Residual ratio $r = |f_{\text{new}}| / |f_{\text{old}}|$

*$\tau = \text{mean of last 3 step ratios}$
Take FP if $r_F < \tau$, else trisect*

NEW f CALLS PER STEP

1 call if FP accepted
3 if rejected (probe still charged)

τ starts at 1; the window never sees future information.

2 Cost-Aware

SIGNAL

Log-progress per function call

*$E = [\ln e_{\text{old}} - \ln e_{\text{new}}] / \text{new calls}$
Pick the higher 3-step mean E*

NEW f CALLS PER STEP

1 call (FP) or 2 calls (trisect)
Bootstrap: FP, then trisection

Greedy: no exploration, so an unused method's estimate can go stale.

3 Stagnation-Aware FP

SIGNAL

Left / right endpoint reuse counters

*If a counter $\geq 3 \rightarrow \text{trisect step}$
Otherwise $\rightarrow \text{false position}$*

NEW f CALLS PER STEP

1 call (FP) or 2 calls (forced trisection)

Reuse limit 2 also tested; counters track real endpoint reuse.

Experimental Setup

15

benchmark equations

Table 1 of the base paper, same intervals

10

methods

5 classical · 2 static hybrids · 3 adaptive

10^{-14}

residual tolerance

stop only when $|f(x)| \leq \varepsilon$; max 1,000 iterations

10

timed runs per pair

after 1 warm-up; method order shuffled (seed 402)

80

digit reference roots

independent mpmath check of every root

5

stress equations

Each one probes a failure mode the paper's equations don't

FAIR ACCOUNTING

- Every distinct $f(x)$ call counts, including initial endpoints and rejected probes. A cached point is never charged twice.
- Failures stay in the totals. Pairwise comparisons use only problems where both methods succeed inside the interval.
- Newton's f' calls are reported separately, not folded into f -call rankings.

Step 1: Reproducing the Base Paper

Static BF

122

total iterations (paper: 121)

Static TF

99

total iterations (paper: 98)

28 of 30

per-problem iteration counts match the paper exactly.

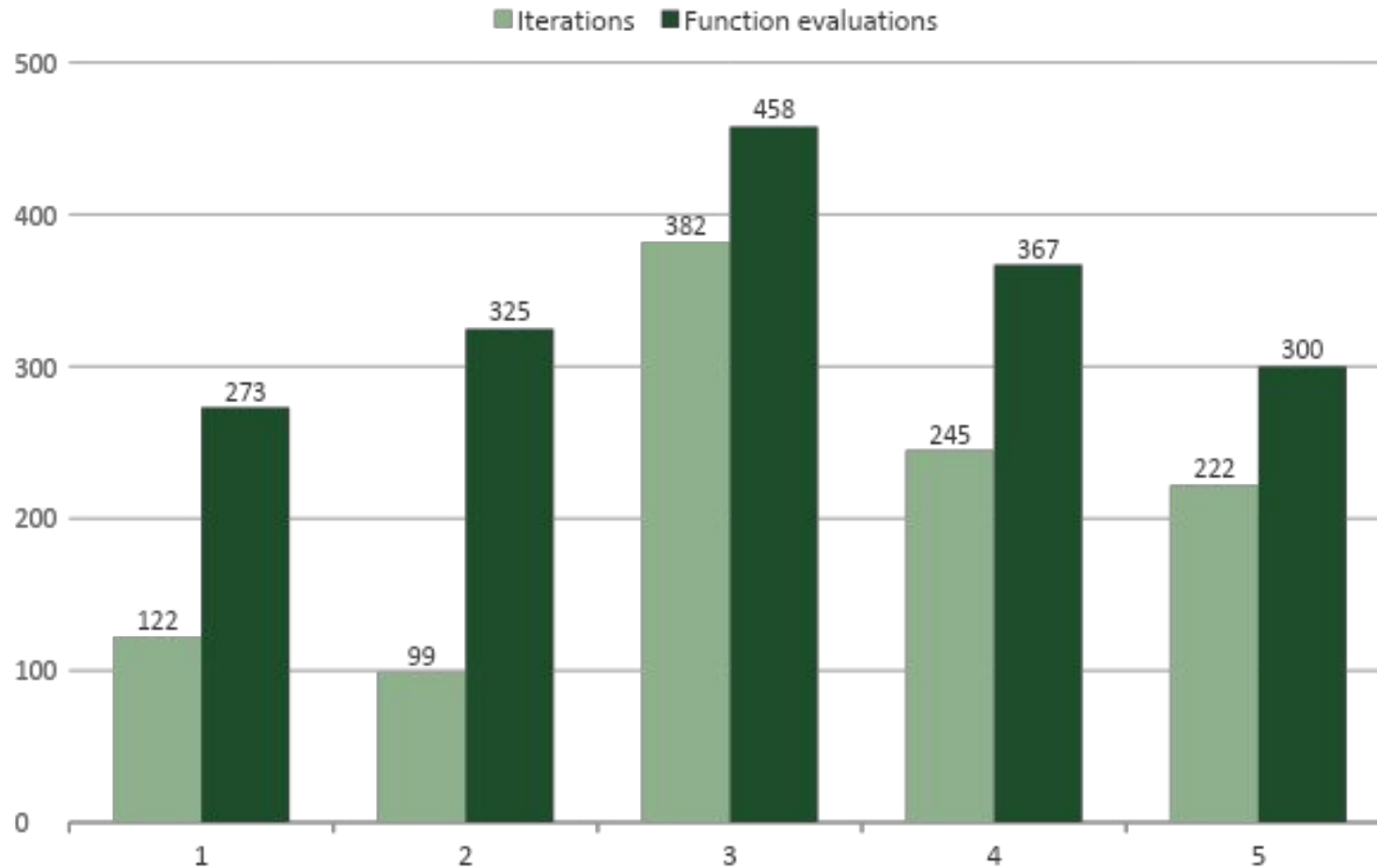
The only difference is P3 ($x^2 - 10$), +1 iteration for both hybrids, most likely from floating-point ordering in the false-position formula.

WHAT WE FOUND IN THE PAPER

- Some printed residuals are labelled 0 but are not zero when the displayed root is substituted back in.
- The text around Figures 4–5 attributes P11's behaviour to P4. P4's root in $[1, 4]$ is 2, found in one trisection step.
- Table 2's "error" is the difference between iterates, but the algorithms stop on $|f(x)|$.
- The $\log_2/\log_3$ step bounds limit interval width, not the residual stopping criterion.

We report published and measured values side by side and never force a match.

Fewer Iterations $\neq$ Less Work



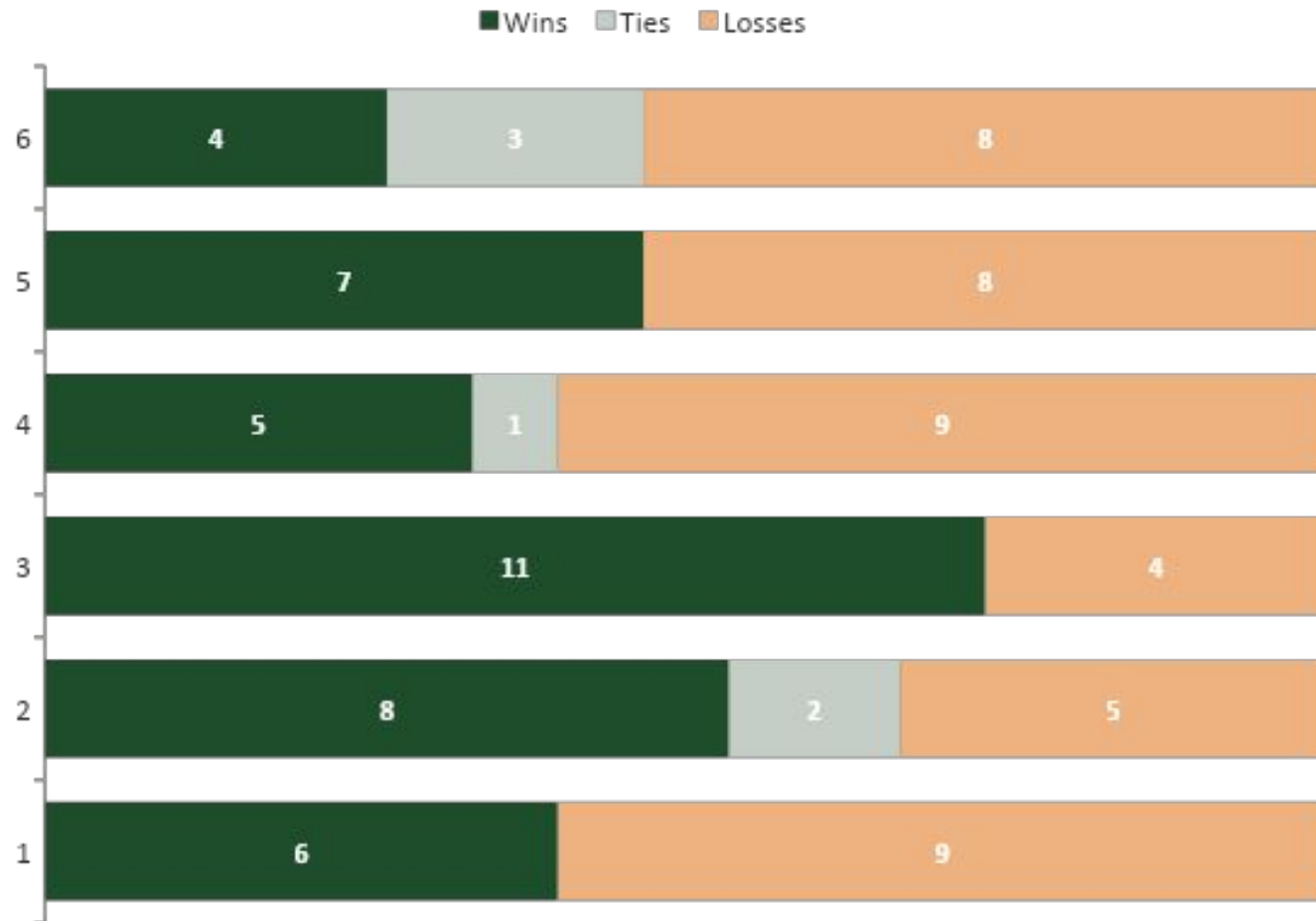
Static TF takes the fewest iterations (99) but spends 325 calls.

Static BF is the cheapest in calls (273). No adaptive policy beats it in aggregate at 10^{-14} .

Stagnation-aware is the best of the three policies: 300 calls, 7.7% fewer than Static TF.

Adaptive threshold is the worst: 458 calls, driven by P11 (98 calls alone).

Does Adaptivity Pay? Head-to-Head



Median f-call ratio (< 1 favours the policy)

Policy	vs TF	vs BF
Threshold	1.118	1.250
Cost-aware	0.824	1.150
Stagnation	0.885	1.038

Adaptivity beats Static TF

Stagnation-aware wins 11 of 15 and cost-aware wins 8 of 15.

...but not Static BF

Every policy loses more often than it wins against BF.

Per-Equation Cost

ID	$f(x)$	Static BF	Static TF	Threshold	Cost-aware	Stagnation
P1	$x^2 - 3$	18	22	16	13	18
P2	$x^2 - 5$	22	26	43	28	25
P3	$x^2 - 10$	18	23	18	15	18
P4	$x^2 - x - 2$	6	5	37	20	24
P5	$x^2 + 2x - 7$	11	22	24	16	19
P6	$x^3 - 2$	20	26	37	59	23
P7	$x \cdot e^x - 7$	24	23	30	16	20
P8	$x - \cos x$	18	23	15	12	15
P9	$x \cdot \sin x - 1$	14	17	10	10	10
P10	$x \cdot \cos x + 1$	22	26	14	61	17
P11	$x^{10} - 1$	26	29	98	44	27
P12	$x^2 + e^{(x/2)} - 5$	18	20	17	13	18
P13	$\sin x \cdot \sinh x + 1$	20	23	40	23	24
P14	$e^x - 3x - 2$	20	23	40	23	24
P15	$\sin x - x^2$	16	17	19	14	18
Total		273	325	458	367	300

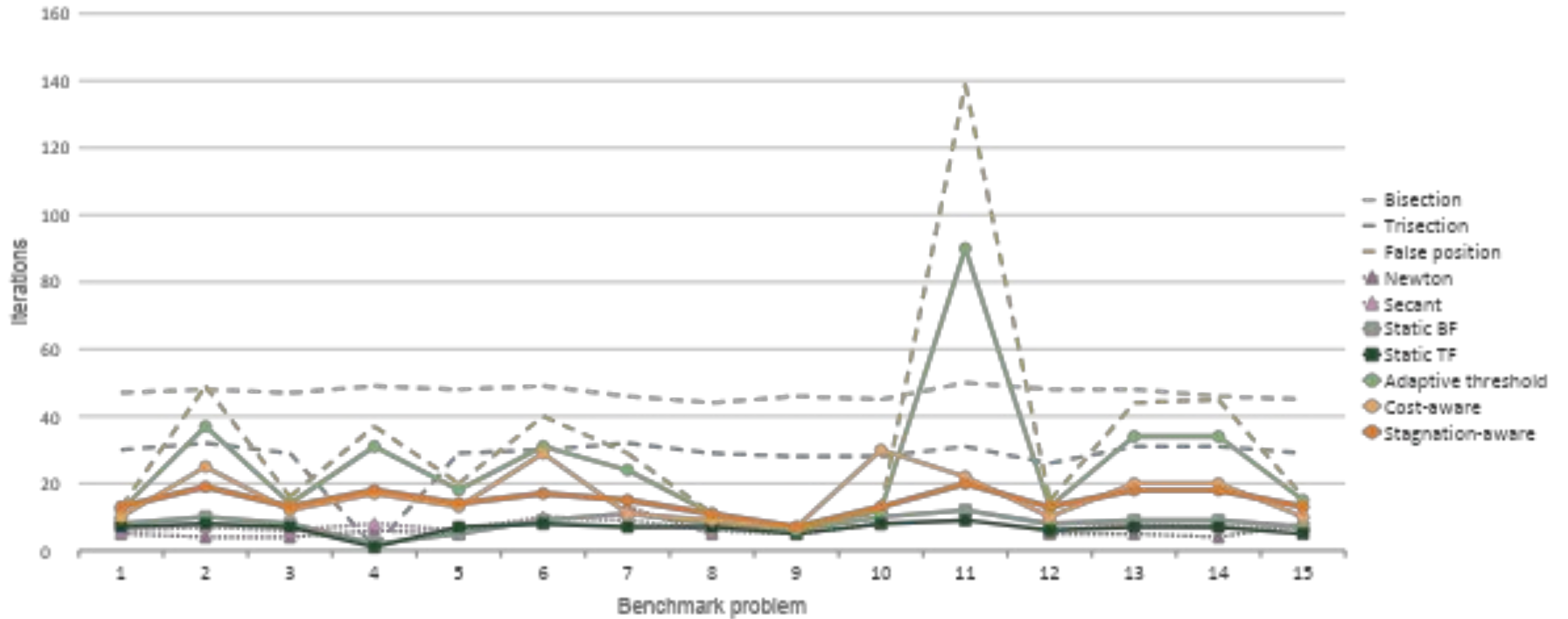
No single winner. Every hybrid is cheapest, or tied for cheapest, on at least one equation.

Threshold outliers. P2, P4, P6, P11, P13 and P14 each cost 37–98 calls (orange).

Cost-aware outliers. P6 (59), P10 (61): after one FP step it trisects 28–29 times. The stale FP estimate is never revisited.

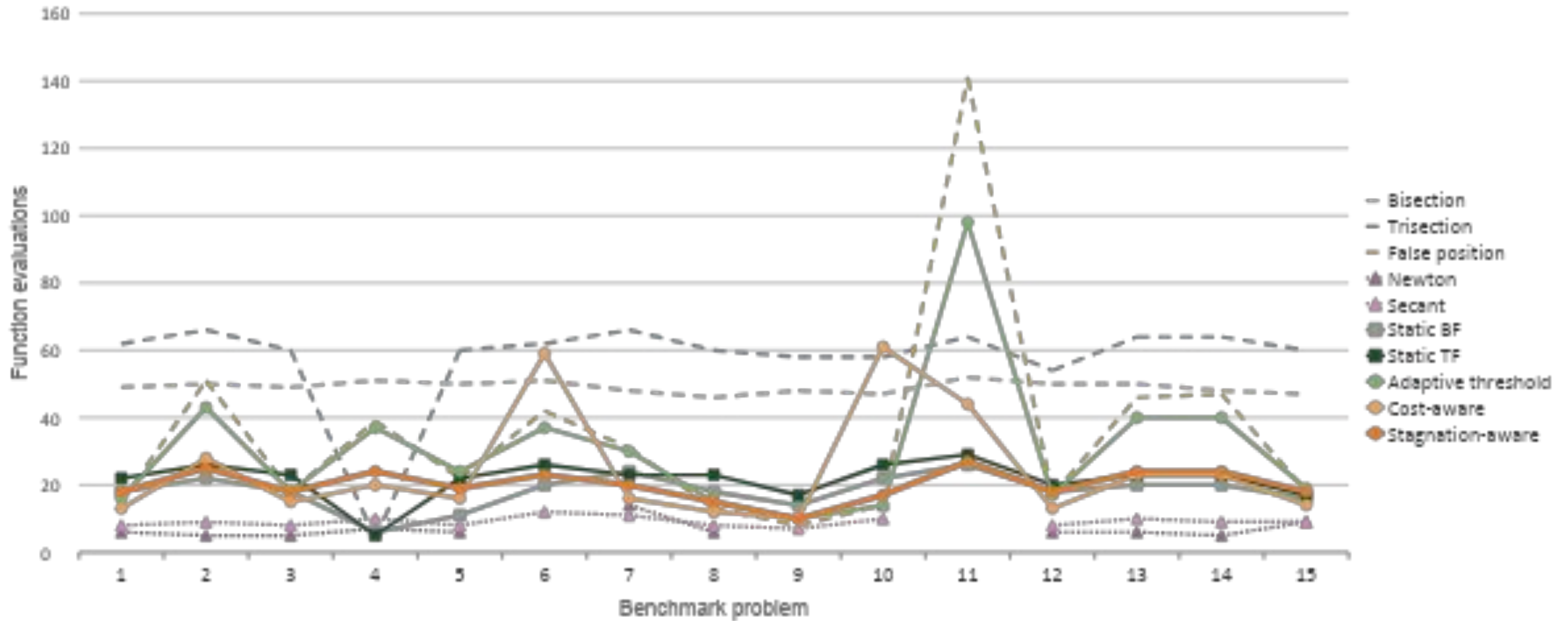
Stagnation-aware never blows up. Its worst case is 27 calls (P11), where the proposal's target failure mode occurs.

Iterations per Problem, All 10 Methods



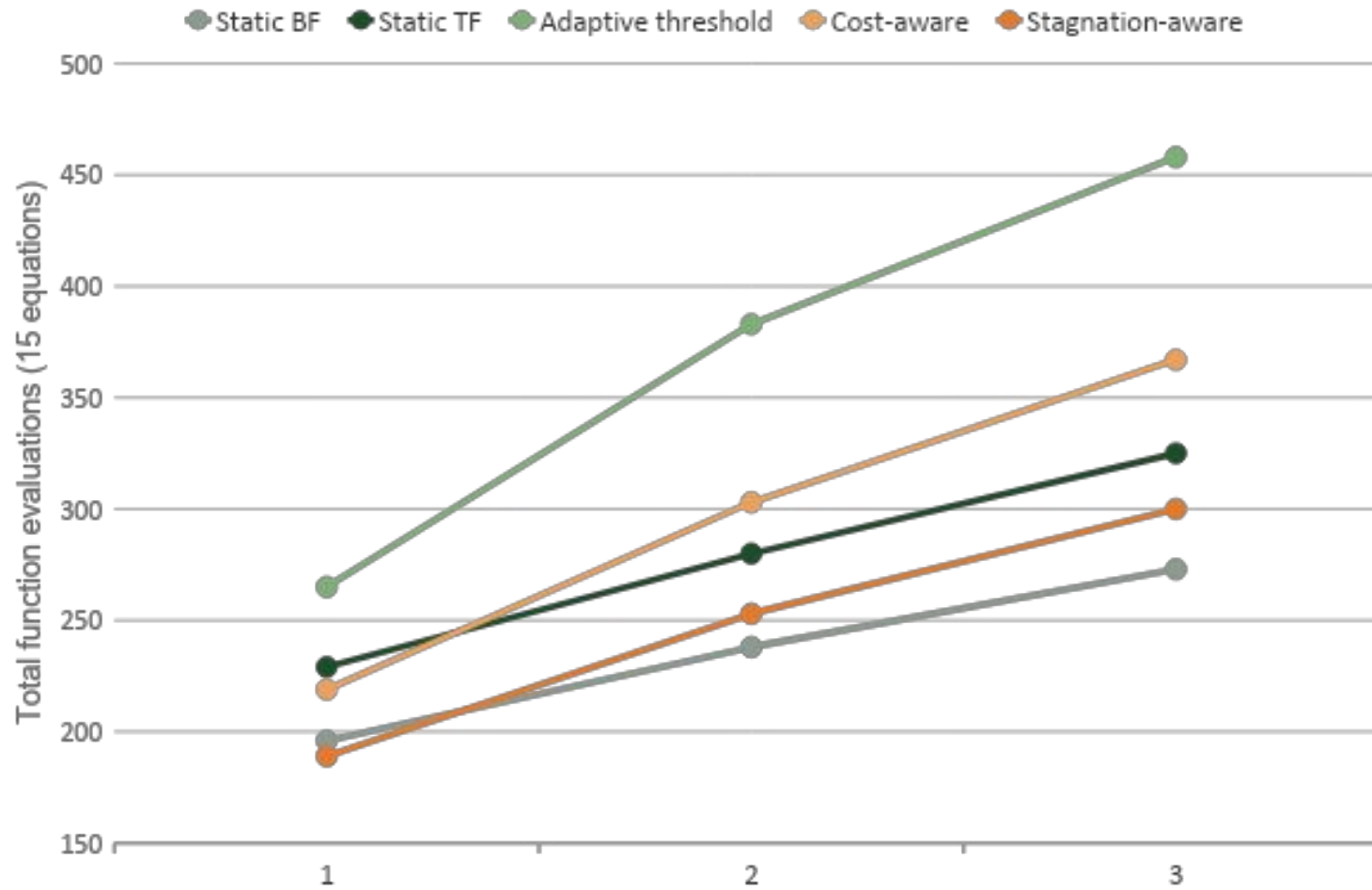
Bisection and trisection are flat because their step count depends only on the interval width. False position and adaptive threshold spike on P11.

Function Evaluations per Problem, All 10 Methods



Stagnation-aware stays within 10–27 calls on every problem; false position ranges from 8 to 141.

Sensitivity: Tolerance and Reuse Limit



Tolerance matters

At 10^{-6} stagnation-aware is the cheapest method overall (189 vs BF's 196). At 10^{-10} and 10^{-14} BF pulls ahead.

Reuse limit 3 $\rightarrow$ 2

Stagnation-aware iterations fall from 222 to 208, but total calls stay at 300. Fewer iterations, same work.

Robust at every ϵ

All three policies solve all 15 equations at each tested tolerance, with no per-equation tuning.

Classical Baselines and Failures

Method	Solved	Iterations	f calls	f' calls
Bisection	15/15	706	736	—
Trisection	15/15	416	862	—
False position	15/15	492	522	—
Newton	12/15	80	92	80
Secant	14/15	104	133	—
Static BF	15/15	122	273	—
Static TF	15/15	99	325	—
Adaptive threshold	15/15	382	458	—
Cost-aware	15/15	245	367	—
Stagnation-aware	15/15	222	300	—

OPEN-METHOD FAILURES

Newton: $f'(a) = 0$ at the left endpoint on P6, P9 and P11. On P10 it converges to a valid root outside $[-2, 4]$.

Secant: stalls on P11 ($x^{10} - 1$ is flat near 0).

STRESS SUITE (5 EXTRA CASES, NOT FROM THE PAPER)

$e^x - 1000$: the best binary64 residual is $\approx 2.3 \times 10^{-13} > \epsilon$, so every bracketed method correctly reports a precision limit.

$x^{20} - 1$, $(x - 0.3)^3$: plain false position hits the 1,000-iteration cap. All hybrids succeed.

Flat cubic, scaled quadratic: a tiny residual does not guarantee a tiny root error, so we also report 80-digit root errors.

Conclusions

Endpoint stagnation is the right signal to adapt on. But adaptivity is not an unconditional win: at tight tolerances, Static BF is still cheaper.

1 Stagnation-aware is the best policy

300 calls vs 325 for Static TF (−7.7%), winning 11 of 15 equations.
Never catastrophic: worst case 27 calls.

2 Residual-ratio thresholds miss endpoint locking

Steady, small improvements keep passing T. On P11 the threshold policy needs 98 calls.

3 Cost-aware looks good per equation, worse in total

Median ratio 0.82 vs TF, but greedy estimates go stale on P6 and P10.

4 Count work, not iterations

Static TF has the fewest iterations yet costs more than BF. Reuse limit 2 cuts iterations, not calls.